Assignment No. 9

Title: Practical Based on String Functions

Objective:

To use and understand various **string functions** in SQL Server using the **AdventureWorks** database.

Theory:

String functions are used to perform operations on character data such as changing case, extracting substrings, trimming spaces, or replacing text. These functions enhance data formatting and help in reporting.

Commonly used string functions in SQL Server:

Function	Description
UPPER(string)	Converts all letters to uppercase
LOWER(string)	Converts all letters to lowercase
LEN(string)	Returns the length of a string
LEFT(string, n)	Returns the leftmost n characters
RIGHT(string, n)	Returns the rightmost n characters
SUBSTRING(string, start, length)	Extracts part of a string
CHARINDEX(substring, string)	Returns the position of a substring
REPLACE(string, old, new)	Replaces a substring with another
LTRIM(string)	Removes leading spaces
RTRIM(string)	Removes trailing spaces
CONCAT(string1, string2, …)	Joins multiple strings
REVERSE(string)	Reverses the order of characters

Table Used:

Person.Person

Column Name	Data Type	Description
BusinessEntityID	INT	Unique ID for each person
FirstName	NVARCHAR(50)	First name of the person
MiddleName	NVARCHAR(50)	Middle name of the person
LastName	NVARCHAR(50)	Last name of the person

Queries:

1) Display all first names in uppercase.

```
SELECT FirstName, UPPER(FirstName) AS UppercaseName
FROM Person.Person;
```

2) Display all last names in lowercase.

```
SELECT LastName, LOWER(LastName) AS LowercaseName
FROM Person.Person;
```

3) Display the length of each first name.

```
SELECT FirstName, LEN(FirstName) AS NameLength
FROM Person.Person;
```

4) Display the first 4 characters of each first name.

```
SELECT FirstName, LEFT(FirstName, 4) AS ShortName
FROM Person.Person;
```

5) Display the last 3 characters of each last name.

```
SELECT LastName, RIGHT(LastName, 3) AS Last3Chars
FROM Person.Person;
```

6) Extract the middle 3 characters from each first name (starting from position 2).

```
SELECT FirstName, SUBSTRING(FirstName, 2, 3) AS ExtractedPart
FROM Person.Person;
```

7) Find the position of letter 'a' in the first name.

```
SELECT FirstName, CHARINDEX('a', FirstName) AS PositionOfA
FROM Person.Person;
```

8) Replace all occurrences of 'a' with '@' in first names.

```
SELECT FirstName, REPLACE(FirstName, 'a', '@') AS ModifiedName
FROM Person.Person;
```

9) Concatenate FirstName and LastName with a space in between.

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName
FROM Person.Person;
```

10) Reverse each person's first name.

```
SELECT FirstName, REVERSE(FirstName) AS ReversedName
FROM Person.Person;
```

Output:

The results display the transformed names based on each string function applied.

Conclusion:

String functions are essential for text manipulation and formatting in SQL Server. They simplify data cleaning and improve data presentation for reports and queries.